

CIS 6270 Practice Exam 1

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Use the following conditions for a timed practice attempt:

- Please write your name on each page. Allow yourself 90 minutes.
- Complete your first attempt individually before consulting the solutions. Record questions to discuss afterward with the instructor and teaching assistants.
- Calculators are not permitted. Work in the space provided and box your final answer for every part. Grade only the boxed final answer.
- Each labeled part is worth 0.5 points and receives either 0 or 0.5 points. There is no partial credit. Each part has one grading target. For multiple choice, box exactly one option number. Algebraically equivalent expressions are accepted.
- You may use a one page cheat sheet (8.5in $\times$ 11in, front page only) containing only material provided in this class. Do not use a calculator, phone, or other device during the timed attempt. Keep your signed cheat sheet with your completed work.

The University of Pennsylvania is a community dedicated to scholarship, leadership, and service and to the principles of honesty, fairness, respect, and accountability. I will use this practice exam to assess my understanding honestly. I will complete the timed attempt using my own work before consulting the solutions or discussing answers.

This practice exam contains 15 points and does not contribute to your course grade.

Name: _____

Signature: _____

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This practice exam is worth 15 points.

Continuous Generation with Fixed Measurements

Suppose you want to complete an image while preserving every observed pixel. Many completions may agree with the available pixels, so you want to generate a distribution of plausible images. More generally, the known measurements can be linear combinations of a continuous state. Every generated sample must reproduce those measurements exactly.

You will develop flow and diffusion models in coordinates that describe the remaining freedom. By reconstructing samples from these coordinates, you can preserve the measurements throughout generation. You must also learn the correct conditional distribution within the permitted set of states. You will examine these two requirements separately in the proofs and experiments.

Figure 1 outlines the proposed construction. You will use the probability, linear algebra, flow matching, score matching, and diffusion tools from class to develop its training objectives and sampling rules.

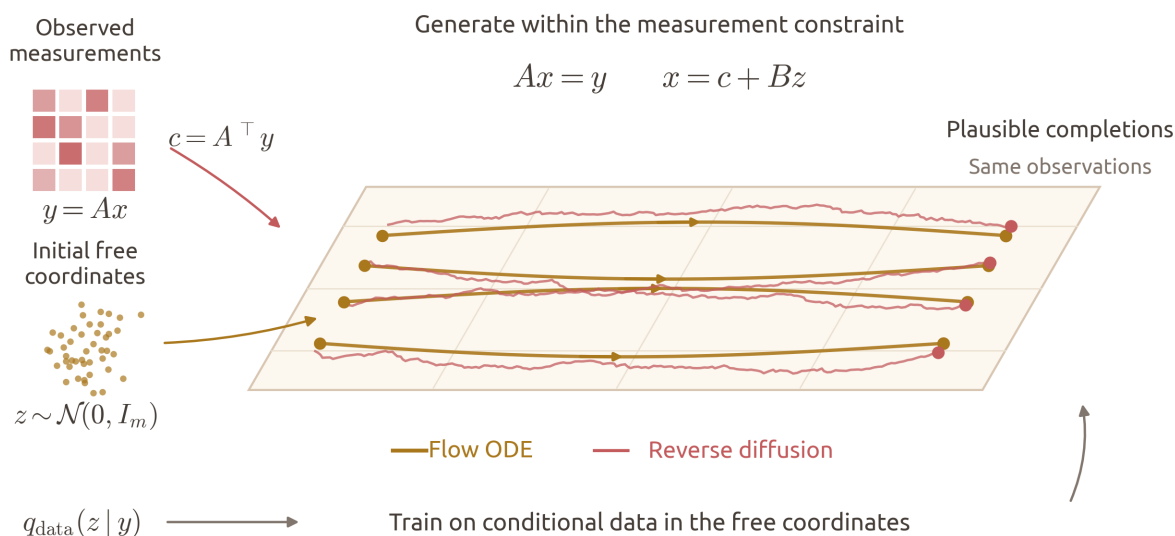


Figure 1. The observed vector y fixes the measured coordinates. The anchor c satisfies those measurements, and the columns of B span the free directions. Train conditional flow or diffusion models for the free coordinates z , then reconstruct states through $x = c + Bz$. Smooth gold paths and irregular red paths illustrate deterministic and stochastic generation within the same permitted set. The points and paths are schematic.

You will build the argument for a methods paper, from its conditional density through training and mathematical guarantees. You will then test implementations, add guidance for competing properties, and interpret hypothetical benchmarks and ablations.

Research brief and notation

The full state is $x \in \mathbb{R}^d$, and the known measurements are $y \in \mathbb{R}^\ell$, with $1 \leq \ell < d$. The fixed measurement matrix $A \in \mathbb{R}^{\ell \times d}$ has orthonormal rows, so $AA^\top = I_\ell$. Define $m = d - \ell$ and choose $B \in \mathbb{R}^{d \times m}$ so that

$$AB = 0, \quad B^\top B = I_m, \quad BB^\top = I_d - A^\top A =: P.$$

The columns of B form an orthonormal basis for the free directions. The matrix P projects onto these directions. You may use all three identities without proof.

Write $c = A^\top y$. Every feasible state has a unique representation

$$x = c + Bz, \quad z = B^\top(x - c) \in \mathbb{R}^m.$$

Thus z records the coordinates that can change while the measurements remain fixed. The square matrix $[A^\top \ B]$ is orthogonal. All densities denoted by q use ordinary m -dimensional volume in z . The data density p_{data} uses d -dimensional volume in x .

Hold y fixed throughout each derivation. Flow time t increases from noise at 0 to data at 1. Diffusion noising time r increases from clean data at 0 to noise at T . Generation reverses this direction. A score is a gradient of log density with respect to the free coordinates. Brownian motions have the dimension stated in each question.

Assume smooth positive densities, finite required moments and integrals, invertible ODE flow maps, vanishing boundary terms, and conditions that justify the stated differentiations and unique solutions. All matrices are constant unless a time dependence is written. Additional constructions are defined where needed.

Step	Points
1. Define the feasible conditional distribution	2.0
2. Train flow and diffusion models	3.0
3. Establish the mathematical guarantees	3.0
4. Test the sampling implementation	3.0
5. Guide competing properties	1.5
6. Interpret benchmarks and ablations	2.5
Total: 30 parts	15.0

Step 1: Defining the Feasible Conditional Distribution (2 points)

Begin by separating the measurements you must preserve from the variation you want to generate. You will need both a reconstruction rule and a density for the remaining coordinates. These calculations determine what the networks should learn and how their outputs become complete samples.

(a) You may obtain a network prediction that disagrees with the observations. To correct that prediction with the smallest possible Euclidean change, find the closest state satisfying the measurements.

For any proposal $\tilde{x} \in \mathbb{R}^d$, solve

$$x^* = \arg \min_{x: Ax=y} \|x - \tilde{x}\|^2.$$

Use the orthogonal decomposition in the research brief. Box one equation for x^* using c , P , and $\tilde{x}$. (0.5 points)

Every feasible candidate is $c + Bz$. Resolve its displacement into orthogonal free and measured components:

$$\|c + Bz - \tilde{x}\|^2 = \|z - B^\top(\tilde{x} - c)\|^2 + \|A(\tilde{x} - c)\|^2.$$

The second term is independent of z . The first has its unique minimum at $z^* = B^\top(\tilde{x} - c)$. Reconstructing gives

$$x^* = c + P(\tilde{x} - c).$$

The correction removes only the component that changes the measurements.

(b) A feasible state can still be an implausible completion. To specify the distribution you want to learn, condition the original data law on the observed measurements.

Let $X \sim p_{\text{data}}$ and suppose the measurement density is positive at y . Apply change of variables to $(AX, B^\top X)$ and then condition on $AX = y$. Box one normalized formula for $q_{\text{data}}(z \mid y)$ using only p_{data} , c , and B . (0.5 points)

The orthogonal coordinate map $x \mapsto (Ax, B^\top x)$ has absolute determinant one. Its joint density is

$$p_{Y,Z}(y, z) = p_{\text{data}}(A^\top y + Bz).$$

Integrate over z to obtain the measurement density, then divide the joint density by that marginal:

$$q_{\text{data}}(z \mid y) = \frac{p_{\text{data}}(c + Bz)}{\int_{\mathbb{R}^m} p_{\text{data}}(c + Bu) du}.$$

The denominator is finite and positive by assumption.

Step 1 continued

(c) You may rescale the free coordinates to improve numerical conditioning. Such a change affects density values, so the paper must specify the volume convention used to report likelihoods.

Let $z = Ru$, where $R \in \mathbb{R}^{m \times m}$ is any invertible matrix. If Z has density q_Z , derive the density q_U of $U = R^{-1}Z$. Box one equation for $q_U(u)$. (0.5 points)

For a measurable region E in u coordinates,

$$\Pr(U \in E) = \int_{RE} q_Z(z) dz = \int_E q_Z(Ru) |\det R| du.$$

The factor $|\det R|$ converts a small volume in u to its volume in z . Therefore

$$\boxed{q_U(u) = q_Z(Ru) |\det R|}.$$

Integrating this expression over u gives one by the same substitution.

(d) Having defined a density for the free coordinates, decide how to describe the law of the reconstructed state. This distinction matters when reporting likelihoods or applying a formula that assumes a density in the full state space.

Let Z have a positive density on $\mathbb{R}^m$ and let $X = c + BZ$. You may use the fact that an m -dimensional affine set has zero d -dimensional volume when $m < d$. Which statement is correct? Choose ONE.

1. $q_Z(B^\top(x - c))$ is a normalized density of X with respect to d -dimensional volume.
2. X has a nonsingular Gaussian density whenever Z is Gaussian.
3. X has no density with respect to d -dimensional volume, but Z has a valid density in the free coordinates.
4. Restricting all samples to $Ax = y$ prevents their probability law from being normalized.

(0.5 points)

The affine set $\{x : Ax = y\}$ has dimension $m < d$ and zero d -dimensional volume. Yet X belongs to that set with probability one. An ordinary d -dimensional density would assign probability zero to every set of zero volume, which contradicts this probability-one event.

Normalization remains valid through $\int q_Z(z) dz = 1$. Thus

$$\boxed{3}$$

Likelihoods for this construction must use the stated free-coordinate volume.

Step 2: Training Flow and Diffusion Models (3 points)

You now have a conditional data distribution in m free coordinates. Train each model with the observed value y available as an input. For flow matching, draw $Z_0 \sim \mathcal{N}(0, I_m)$ and $Z_1 \sim q_{\text{data}}(\cdot | y)$ independently, then use

$$Z_t = (1 - t)Z_0 + tZ_1, \quad U = Z_1 - Z_0.$$

Here U is the sampled velocity target. The corresponding full state is $X_t = c + BZ_t$. These training pairs remain feasible at every time.

(a) One implementation predicts a full d -dimensional velocity before projecting it. To understand how much of its regression error affects generation, separate mistakes in the free directions from predictions that change the measurements.

Let $w = w_\theta(X_t, t, y) \in \mathbb{R}^d$ be the full velocity prediction. Derive an orthogonal decomposition of its sampled flow-matching loss $\|w - BU\|^2$. Box one equation whose right-hand side is the sum of a free-coordinate regression error and a measured-coordinate error. (0.5 points)

The orthogonal matrix $[A^\top \ B]$ preserves Euclidean norms. Apply its transpose to the residual:

$$\|w - BU\|^2 = \|B^\top(w - BU)\|^2 + \|A(w - BU)\|^2.$$

Using $B^\top B = I_m$ and $AB = 0$ gives

$$\|w - BU\|^2 = \|B^\top w - U\|^2 + \|Aw\|^2.$$

Only $B^\top w$ determines motion after projection. The second term penalizes a component that projection removes. Predicting the m free coordinates directly avoids allocating outputs to that component.

Step 2 continued

(b) For diffusion training, you can predict either the corruption noise or the score. Before comparing these parameterizations, match their time-dependent loss weights so that an apparent training improvement does not come from changing the objective.

At noising time r , use $Z_r = a_r Z_0 + \sigma_r \varepsilon$. Here $Z_0 \sim q_{\text{data}}(\cdot \mid y)$ is the clean sample, $\varepsilon \sim \mathcal{N}(0, I_m)$ is independent noise, $a_r > 0$, and $\sigma_r > 0$. The conditional score target is $G_r = -\varepsilon/\sigma_r$. Parameterize $s_\psi = -\varepsilon_\psi/\sigma_r$. Find the weight $w(r)$ such that

$$w(r) \|s_\psi(Z_r, r, y) - G_r\|^2 = \|\varepsilon_\psi(Z_r, r, y) - \varepsilon\|^2$$

for every training example. Box only $w(r)$. (0.5 points)

Substitute both score expressions into the residual:

$$s_\psi - G_r = -\frac{\varepsilon_\psi - \varepsilon}{\sigma_r}, \quad \|s_\psi - G_r\|^2 = \frac{\|\varepsilon_\psi - \varepsilon\|^2}{\sigma_r^2}.$$

The equality must hold for every residual, so $w(r)/\sigma_r^2 = 1$. Hence

$$\boxed{w(r) = \sigma_r^2.}$$

The two losses agree sample by sample with this weight.

(c) A sliced score-matching calculation uses derivatives along random directions. In the constrained model, those directions must correspond to variation in the free coordinates. Determine which trace a projected random probe estimates.

Fix x and a differentiable vector field $w : \mathbb{R}^d \rightarrow \mathbb{R}^d$. Its Jacobian is $J_w(x)$, with entry (i, j) equal to $\partial w_i / \partial x_j$. Draw $\xi \sim \mathcal{N}(0, I_m)$ and set $\eta = B\xi$. Box one trace expression for $\mathbb{E}_\xi[\eta^\top J_w(x) \eta]$. (0.5 points)

Write $K = B^\top J_w(x) B$. For the standard Gaussian probe, $\mathbb{E}[\xi_i \xi_j] = \delta_{ij}$, where δ_{ij} is one when $i = j$ and zero otherwise. Therefore

$$\mathbb{E}[\xi^\top K \xi] = \sum_{i,j} K_{ij} \mathbb{E}[\xi_i \xi_j] = \sum_i K_{ii}.$$

Consequently,

$$\boxed{\mathbb{E}_\xi[\eta^\top J_w(x) \eta] = \text{tr}(B^\top J_w(x) B).}$$

This is the divergence in z of $B^\top w(c + Bz)$, by the chain rule.

Step 2 continued

(d) To compare diffusion samplers at different resolutions, you need the forward transition between any two retained noise levels. Derive this transition by composing the original noising steps, so skipped steps retain the intended corruption distribution.

In free coordinates, a DDPM forward chain obeys

$$Z_i = \sqrt{\alpha_i} Z_{i-1} + \sqrt{1 - \alpha_i} \varepsilon_i, \quad \bar{\alpha}_i = \prod_{n=1}^i \alpha_n,$$

where $0 < \alpha_i < 1$ and the ε_i are independent standard Gaussian vectors in $\mathbb{R}^m$. For $0 \leq j < k$, derive $q(z_k | z_j)$. Use $\bar{\alpha}_0 = 1$ and box one Gaussian distribution, including its mean and covariance. (0.5 points)

Unroll the recursion from j to k :

$$Z_k = \left(\prod_{n=j+1}^k \sqrt{\alpha_n} \right) Z_j + \sum_{i=j+1}^k \sqrt{1 - \alpha_i} \left(\prod_{n=i+1}^k \sqrt{\alpha_n} \right) \varepsilon_i.$$

The independent noises have zero mean. Their covariance coefficient is

$$\sum_{i=j+1}^k (1 - \alpha_i) \prod_{n=i+1}^k \alpha_n = 1 - \prod_{n=j+1}^k \alpha_n,$$

because each summand is the product difference

$$(1 - \alpha_i) \prod_{n=i+1}^k \alpha_n = \prod_{n=i+1}^k \alpha_n - \prod_{n=i}^k \alpha_n.$$

Successive terms cancel. The signal coefficient is the square root of the remaining product. Thus

$$q(z_k | z_j) = \mathcal{N} \left(\sqrt{\frac{\bar{\alpha}_k}{\bar{\alpha}_j}} z_j, \left(1 - \frac{\bar{\alpha}_k}{\bar{\alpha}_j} \right) I_m \right).$$

This transition depends on the cumulative noise levels at the retained endpoints.

Step 2 continued

(e) The skipped forward transition specifies the noise levels. To generate with a deterministic DDIM step, estimate a clean state at the current level and reuse the same predicted noise at the next retained level. Express this update directly in quantities available from one network evaluation.

Let $j < k$ and $e_k = \varepsilon_\psi(z_k, k, y)$. Use

$$\hat{z}_0 = \frac{z_k - \sqrt{1 - \bar{\alpha}_k} e_k}{\sqrt{\bar{\alpha}_k}}, \quad z_j = \sqrt{\bar{\alpha}_j} \hat{z}_0 + \sqrt{1 - \bar{\alpha}_j} e_k.$$

Substitute and collect terms. Box one equation for z_j using only z_k , e_k , $\bar{\alpha}_j$, and $\bar{\alpha}_k$.
(0.5 points)

Substitute the clean-state estimate into the update:

$$z_j = \sqrt{\frac{\bar{\alpha}_j}{\bar{\alpha}_k}} (z_k - \sqrt{1 - \bar{\alpha}_k} e_k) + \sqrt{1 - \bar{\alpha}_j} e_k.$$

Separate the coefficient of the current state from the coefficient of predicted noise:

$$z_j = \sqrt{\frac{\bar{\alpha}_j}{\bar{\alpha}_k}} z_k + \left(\sqrt{1 - \bar{\alpha}_j} - \sqrt{\frac{\bar{\alpha}_j}{\bar{\alpha}_k}} \sqrt{1 - \bar{\alpha}_k} \right) e_k.$$

Reconstruct $x_j = c + Bz_j$ after the step. The observed anchor c stays fixed. Finite-step accuracy still depends on the learned predictor and the retained noise levels.

Step 2 continued

(f) You can also analyze the diffusion model through a continuous SDE. Derive that limit in the full state coordinates to identify exactly where its drift and Brownian fluctuations act.

Set $\alpha_i = 1 - \beta(r_i)\Delta r$ in the forward chain, where $\beta(r) > 0$ is the continuous noising rate. Use $\sqrt{1-u} = 1 - u/2 + o(u)$ and $X_r = c + BZ_r$. Box the limiting forward SDE for X_r using P , c , B , and standard m -dimensional Brownian motion W_r . (0.5 points)

One free-coordinate increment is

$$Z_{r+\Delta r} - Z_r = -\frac{1}{2}\beta(r)Z_r\Delta r + \sqrt{\beta(r)}\sqrt{\Delta r}\varepsilon + o(\Delta r).$$

The Gaussian increment $\sqrt{\Delta r}\varepsilon$ becomes a Brownian increment. Since c and B are constant, $dX_r = B dZ_r$. For a feasible state, $BZ_r = X_r - c = P(X_r - c)$. Therefore

$$\boxed{dX_r = -\frac{1}{2}\beta(r)P(X_r - c)dr + \sqrt{\beta(r)}B dW_r.}$$

The noise covariance per unit time is $\beta(r)BB^\top = \beta(r)P$. Both drift and noise act in the free directions.

Step 3: Establishing the Mathematical Guarantees (3 points)

The training and sampling equations now specify how to generate in the free coordinates. The paper must establish which properties follow from this construction. You will prove preservation of measurements, examine how diffusion changes uncertainty, and derive checks on the learned conditional score.

(a) Begin with the observation constraint itself. A stochastic path receives infinitely many random increments, so a guarantee should apply along the whole path. Check the drift and noise in the measured directions to establish this claim.

Consider the general lifted dynamics

$$dX_t = Bb_t(B^\top(X_t - c)) dt + g(t)B dW_t, \quad AX_0 = y,$$

where $b_t : \mathbb{R}^m \rightarrow \mathbb{R}^m$ is a drift, $g(t)$ is a scalar, and W_t is standard m -dimensional Brownian motion. Apply Itô's lemma to the linear map $x \mapsto Ax$. Box the resulting pathwise identity for AX_t . The case $g = 0$ includes a flow ODE. (0.5 points)

The Jacobian of $x \mapsto Ax$ is A and its Hessian is zero. Itô's lemma therefore gives

$$d(AX_t) = ABb_t(B^\top(X_t - c)) dt + g(t)AB dW_t = 0,$$

using $AB = 0$ in both terms. Integrate from 0 to t and use the feasible initial state:

$$AX_t = AX_0 + \int_0^t 0 ds + \int_0^t 0 dW_s.$$

Consequently,

$$\boxed{AX_t = y \text{ for every } t, \text{ almost surely.}}$$

With continuous sample paths, this identity holds simultaneously over the time interval. It also holds for the deterministic case.

Step 3 continued

(b) Although the measurements stay fixed, diffusion can still increase uncertainty among their possible completions. Quantify this effect within the free coordinates using the entropy and density-evolution tools from class.

For this part, isolate pure noising, $dZ_r = g(r)dW_r$. Its density satisfies

$$\partial_r q_r = \frac{1}{2}g(r)^2 \Delta_z q_r.$$

Define the differential entropy $H(q_r) = -\int q_r(z) \log q_r(z) dz$ and score $s_r = \nabla_z \log q_r$. Differentiate the entropy, use probability conservation, and integrate by parts. Box one expression for $\frac{d}{dr}H(q_r)$ in terms of $g(r)$ and an expectation of the score. (0.5 points)

Differentiate under the integral:

$$\frac{dH}{dr} = -\int (1 + \log q_r) \partial_r q_r dz.$$

The boundary assumption gives $\int \partial_r q_r dz = \frac{1}{2}g(r)^2 \int \Delta q_r dz = 0$, which removes the term without $\log q_r$. Substituting the heat equation leaves

$$\frac{dH}{dr} = -\frac{g(r)^2}{2} \int \log q_r \Delta q_r dz.$$

Integration by parts removes the Laplacian:

$$-\int \log q_r \Delta q_r dz = \int \nabla \log q_r \cdot \nabla q_r dz = \int q_r \|s_r\|^2 dz.$$

Thus

$$\boxed{\frac{d}{dr}H(q_r) = \frac{g(r)^2}{2} \mathbb{E}_{q_r} \|s_r(Z_r)\|^2.}$$

The right-hand side is nonnegative. This statement concerns pure noising in free-coordinate volume, with the drift set to zero.

Step 3 continued

(c) The physical set of valid completions does not depend on which orthonormal basis you use for its free directions. To make the generative construction consistent with that fact, determine how its drift must transform when you rotate the basis.

Let $\tilde{B} = BO$, where $O \in \mathbb{R}^{m \times m}$ is orthogonal. The same full state has coordinates $\tilde{Z} = O^\top Z$. Suppose

$$dZ_t = b_t(Z_t)dt + g(t)dW_t.$$

Derive the drift $\tilde{b}_t$ required in the rotated coordinates. You may use the fact that an orthogonal transform of standard Brownian motion is standard. For the ODE case $g = 0$, also verify that $\tilde{q}_t(u) = q_t(Ou)$ satisfies the continuity equation whenever q_t does. Box only $\tilde{b}_t(u)$. (0.5 points)

Apply the constant linear transformation:

$$d\tilde{Z}_t = O^\top b_t(O\tilde{Z}_t)dt + g(t)O^\top dW_t.$$

The process $\tilde{W}_t = O^\top W_t$ has continuous paths and independent Gaussian increments. Their covariance is

$$\text{Cov}(\tilde{W}_{t+h} - \tilde{W}_t) = hO^\top O = hI_m,$$

so $\tilde{W}$ is standard Brownian motion. Hence the required drift is

$$\boxed{\tilde{b}_t(u) = O^\top b_t(Ou).}$$

Couple the two processes by this rotated Brownian motion and transformed initial state. Their full states agree:

$$c + \tilde{B}\tilde{Z}_t = c + BOO^\top Z_t = c + BZ_t.$$

For the ODE case, the Jacobian of the transformed flux is $O^\top J_{q_t b_t}(Ou)O$, where $J_{q_t b_t}$ is the Jacobian of the vector field $q_t b_t$. The trace identity gives

$$\nabla_u \cdot (\tilde{q}_t \tilde{b}_t)(u) = \text{tr}(O^\top J_{q_t b_t}(Ou)O) = \nabla_z \cdot (q_t b_t)(Ou).$$

Also $\partial_t \tilde{q}_t(u) = \partial_t q_t(Ou)$. Thus $\partial_t \tilde{q}_t + \nabla_u \cdot (\tilde{q}_t \tilde{b}_t) = 0$. The rotated coordinates preserve the transported probability law.

Step 3 continued

(d) As an alternative, you can allow a small measurement error and add a strong penalty. Derive the resulting score to see how the penalty affects the full-state dynamics as the allowed error approaches zero.

For a fixed free-coordinate density q with score $s = \nabla_z \log q$, define the normalized full-state density

$$p_\epsilon(x) = q(B^\top(x - c)) (2\pi\epsilon^2)^{-\ell/2} \exp\left[-\frac{\|Ax - y\|^2}{2\epsilon^2}\right], \quad \epsilon > 0.$$

Box one equation for $\nabla_x \log p_\epsilon(x)$. (0.5 points)

Take the logarithm and differentiate each state-dependent term:

$$\nabla_x \log q(B^\top(x - c)) = Bs(B^\top(x - c)),$$

$$\nabla_x \frac{\|Ax - y\|^2}{2\epsilon^2} = \epsilon^{-2} A^\top(Ax - y).$$

The normalization factor has zero spatial derivative. Therefore

$$\nabla_x \log p_\epsilon(x) = Bs(B^\top(x - c)) - \epsilon^{-2} A^\top(Ax - y).$$

The measured-direction restoring strength grows as ϵ^{-2} .

(e) Exact measurements alone cannot verify a learned conditional score. Construct a sample-based identity that tests the score against a chosen smooth vector field, so later experiments can check density information within the feasible set.

Let $h : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be smooth with compact support, and let $s = \nabla \log q$ be the unknown true score. Define

$$\mathcal{R}_\psi(h) = \mathbb{E}_q[(s_\psi(Z) - s(Z))^\top h(Z)].$$

Use integration by parts to remove s from this expression. Box one formula for $\mathcal{R}_\psi(h)$ using only samples from q , the learned score s_ψ , and derivatives of h . (0.5 points)

Since $qs = \nabla q$, sum integration by parts over coordinates:

$$\mathbb{E}_q[s^\top h] = \sum_i \int h_i \partial_i q \, dz = - \sum_i \int q \partial_i h_i \, dz = -\mathbb{E}_q[\nabla \cdot h].$$

Substitute this identity into the definition of the residual:

$$\mathcal{R}_\psi(h) = \mathbb{E}_q[s_\psi(Z)^\top h(Z) + \nabla \cdot h(Z)].$$

An exact score has residual zero for every admissible test field.

Step 3 continued

(f) To use the residual responsibly, distinguish the population theorem from a finite diagnostic. Establish what vanishing residuals can certify before interpreting a small set of empirical checks in the experiments.

Assume $q > 0$ and the score error $s_\psi - s$ is continuous. Which statement follows from the identity in (e)? Choose ONE.

1. Zero residual for one nonzero test field proves that the learned score is exact everywhere.
2. Zero population residual for every smooth compactly supported test field implies $s_\psi = s$. A finite collection of zero residuals alone does not establish this conclusion.
3. Enforcing $AX = y$ makes every score residual zero, regardless of the learned field.
4. Zero residuals imply that all generated samples equal a single conditional mean.

Support your selection by considering a test field aligned with a nonzero local score error. Box only the option number. (0.5 points)

Write $e = s_\psi - s$. If $e(z_0) \neq 0$ at some point, set $a = e(z_0)$. Continuity gives a small neighborhood in which $e(z)^\top a > 0$.

Choose a nonnegative smooth function φ with compact support in that neighborhood, positive somewhere, and use $h(z) = a\varphi(z)$. Then

$$\mathcal{R}_\psi(h) = \int q(z)e(z)^\top a \varphi(z) \, dz > 0,$$

because $q > 0$. This contradicts the assumption of zero residual for every test field. Thus the error must vanish everywhere under the stated continuity assumptions.

A finite family tests only finitely many averages. A nonzero error can have zero averages against those particular fields. The correct selection is

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Step 4: Testing the Sampling Implementation (3 points)

The preceding guarantees apply to the specified equations. Numerical updates can violate them if noise, projection, or initialization is implemented inconsistently. You will derive diagnostic predictions for these implementation choices before examining experimental results.

(a) First check the noise dimension. A common implementation projects the drift but adds an ordinary full-state Gaussian increment. Quantify how much that single step changes the supposedly fixed measurements.

Start from $Ax = y$ and use

$$x^+ = x + hBb + g\sqrt{h}\xi, \quad \xi \sim \mathcal{N}(0, I_d),$$

where $b \in \mathbb{R}^m$ is fixed during the step, $h > 0$, and $g > 0$. Box one equation for $\mathbb{E}_\xi \|Ax^+ - y\|^2$. (0.5 points)

Use $Ax = y$ and $AB = 0$ to obtain $Ax^+ - y = g\sqrt{h}A\xi$. Then

$$\mathbb{E}\|A\xi\|^2 = \text{tr}(A\mathbb{E}[\xi\xi^\top]A^\top) = \text{tr}(AA^\top) = \ell.$$

Therefore

$$\boxed{\mathbb{E}_\xi \|Ax^+ - y\|^2 = g^2 h \ell.}$$

Projecting the drift leaves this stochastic measurement error intact.

(b) You could replace exact projection with the soft penalty from Step 3(d). A stronger penalty pulls samples toward the observations more quickly, but an explicit numerical step must resolve that faster motion.

Isolate the deterministic penalty dynamics

$$\dot{x} = -\epsilon^{-2}A^\top(Ax - y).$$

Apply explicit Euler with step h . Determine the condition on h for every nonzero measurement error to decay under repeated steps. Box the open interval of admissible step sizes. (0.5 points)

Let $e_n = Ax_n - y$. The Euler step and $AA^\top = I_\ell$ give

$$e_{n+1} = e_n - h\epsilon^{-2}AA^\top e_n = (1 - h/\epsilon^2)e_n.$$

Every nonzero error decays exactly when $|1 - h/\epsilon^2| < 1$. Solving this inequality yields

$$\boxed{0 < h < 2\epsilon^2.}$$

At the upper endpoint the error alternates without decay. Larger steps amplify it.

Step 4 continued

(c) Projecting only the final state guarantees that the reported sample satisfies the observations. Intermediate violations can still affect its free coordinates when the network couples measured and free directions. Compare two projection schedules to quantify that effect.

Let the centered state be $r_0 = x - c$ with $Pr_0 = r_0$, and let the full-state velocity be $w(x) = M(x - c)$ for an arbitrary $d \times d$ matrix M . Take two Euler steps of size h . Write

$$r_{\text{end}} = P(I_d + hM)^2 r_0, \quad r_{\text{each}} = P(I_d + hM)P(I_d + hM)r_0.$$

The first scheme projects after both steps. The second projects after each step. Expand both expressions and box a simplified formula for $r_{\text{end}} - r_{\text{each}}$ that displays the measured-direction coupling. (0.5 points)

Expand the two-step map with final projection:

$$r_{\text{end}} = r_0 + 2hPMr_0 + h^2PM^2r_0.$$

For projection after each step, use $Pr_0 = r_0$ and $P^2 = P$:

$$r_{\text{each}} = r_0 + 2hPMr_0 + h^2PMPMr_0.$$

Subtract and factor the quadratic term:

$$\boxed{r_{\text{end}} - r_{\text{each}} = h^2PM(I_d - P)Mr_0.}$$

The first velocity evaluation can create a measured-direction displacement $(I_d - P)Mr_0$. At the next evaluation, PM can convert that displacement into a free-coordinate change. Final projection preserves that change even though it removes the final measurement error.

Step 4 continued

(d) Even with exact dynamics and projection, a mismatched starting distribution can produce incorrect completions. Determine whether integrating an exact probability-flow ODE more accurately can remove this initialization error.

Let $\Phi : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be the invertible map of an exact generation ODE. It sends the intended noisy density q_T to the intended clean density q_0 . Start the same map from another positive density ρ_T , obtaining ρ_0 . Using

$$D_{\text{KL}}(\rho \| q) = \int \rho(z) \log \frac{\rho(z)}{q(z)} \, dz,$$

prove how the divergence changes under the shared map. Box an equality between the terminal divergence and its value at initialization. (0.5 points)

At corresponding points $z = \Phi(u)$, change of variables gives

$$\rho_0(\Phi(u)) = \frac{\rho_T(u)}{|\det D\Phi(u)|}, \quad q_0(\Phi(u)) = \frac{q_T(u)}{|\det D\Phi(u)|}.$$

The common determinant cancels from the density ratio. Substitute $z = \Phi(u)$ into the terminal divergence:

$$\begin{aligned} D_{\text{KL}}(\rho_0 \| q_0) &= \int \frac{\rho_T(u)}{|\det D\Phi(u)|} \log \frac{\rho_T(u)}{q_T(u)} |\det D\Phi(u)| \, du \\ &= \int \rho_T(u) \log \frac{\rho_T(u)}{q_T(u)} \, du. \end{aligned}$$

Therefore

$$\boxed{D_{\text{KL}}(\rho_0 \| q_0) = D_{\text{KL}}(\rho_T \| q_T)}.$$

Refining the numerical approximation to this exact invertible map cannot remove a fixed initialization mismatch. This invariance statement concerns the shared deterministic map.

Step 4 continued

(e) Use a solvable conditional distribution to make the initialization issue concrete. A finite noising horizon can retain information about the observed sample, even when the noisy covariance already equals the identity.

Suppose $q_0 = \mathcal{N}(\mu, I_m)$ and the variance-preserving noising law is $q_r = \mathcal{N}(a_r\mu, I_m)$, with $a_0 = 1$ and $\dot{a}_r = -\beta(r)a_r/2$. The exact probability-flow velocity in noising time is

$$u_r(z) = -\frac{1}{2}\beta(r)(z + \nabla_z \log q_r(z)).$$

Integrate this ODE backward from T to 0, but initialize $Z_T \sim \mathcal{N}(0, I_m)$. Box the resulting terminal mean bias $\mathbb{E}[Z_0] - \mu$. (0.5 points)

The exact score is $-(z - a_r\mu)$. Hence

$$u_r(z) = -\frac{1}{2}\beta(r)a_r\mu = \dot{a}_r\mu.$$

Backward integration gives $Z_0 = Z_T + \int_T^0 \dot{a}_r\mu \, dr = Z_T + (1 - a_T)\mu$. Since the implemented initialization has mean zero,

$$\boxed{\mathbb{E}[Z_0] - \mu = -a_T\mu.}$$

The bias remains under exact integration whenever $a_T\mu \neq 0$.

(f) The initialization law also depends on how noise is represented after rescaling coordinates. Return to $z = Ru$ from Step 1(c) and determine the covariance required for an isotropic Brownian increment in the physical free coordinates.

Suppose a numerical Gaussian noise increment in u has covariance g^2hC_U , where $g, h > 0$. Require the corresponding increment in z to have covariance g^2hI_m . Box the unique matrix C_U in terms of the invertible matrix R . (0.5 points)

Linear covariance transformation gives

$$\text{Cov}(\Delta Z) = R \text{Cov}(\Delta U) R^\top = g^2hRC_U R^\top.$$

Match this with g^2hI_m and multiply by the two inverses:

$$\boxed{C_U = R^{-1}R^{-\top}.$$

For example, $\Delta U = g\sqrt{h}R^{-1}\xi$ with $\xi \sim \mathcal{N}(0, I_m)$ has exactly this covariance.

Step 5: Guiding Competing Properties (1.5 points)

Feasible completions may differ in properties that matter to the user. Add guidance within the free coordinates so the measurements remain fixed. You will account for the dependence of a predicted clean sample on the noisy state, then examine competing objectives and joint conditioning.

(a) A property predictor may be reliable only on completed samples. To guide an intermediate noisy state, evaluate the property on a denoised estimate and differentiate through that estimate. Keeping this dependency is necessary to compute the intended guidance direction.

At a fixed noising time r , define

$$\hat{X}_0(z) = c + \frac{B}{a_r}(z + \sigma_r^2 s_\psi(z, r, y)), \quad a_r > 0.$$

Let $F : \mathbb{R}^d \rightarrow \mathbb{R}$ be a differentiable property to maximize. Let $J_\psi = \partial s_\psi / \partial z$ use the Jacobian convention from Step 2(c). Box one equation for $\nabla_z F(\hat{X}_0(z))$, including all chain-rule factors. (0.5 points)

At the fixed time, a_r and σ_r are constants. The Jacobian of the estimated full state is

$$D_z \hat{X}_0 = \frac{B}{a_r}(I_m + \sigma_r^2 J_\psi).$$

For a scalar property, the input gradient is the transpose of this Jacobian times the output gradient:

$$\nabla_z F(\hat{X}_0) = (D_z \hat{X}_0)^\top \nabla_x F(\hat{X}_0).$$

Therefore

$$\nabla_z F(\hat{X}_0(z)) = \frac{1}{a_r}(I_m + \sigma_r^2 J_\psi)^\top B^\top \nabla_x F(\hat{X}_0(z)).$$

Lifting a step along this free-coordinate gradient through B preserves the measurements. Dropping J_ψ changes the derivative being used.

Step 5 continued

(b) Two desirable properties can favor opposing local changes. Before choosing a tradeoff weight, check whether a small guidance step can improve both properties at the current state. This local calculation will help you interpret later multi-objective results.

At one state with $m = 2$, the free-coordinate gradients of two already normalized properties are

$$g_1 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad g_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

Both properties should increase. Use $d_\alpha = \alpha g_1 + (1 - \alpha)g_2$, with $0 \leq \alpha \leq 1$. Derive the weights for which both directional derivatives $g_1^\top d_\alpha$ and $g_2^\top d_\alpha$ are strictly positive. Box one open interval for α . (0.5 points)

Form the weighted direction:

$$d_\alpha = \begin{pmatrix} 3\alpha - 1 \\ 1 - \alpha \end{pmatrix}.$$

The first property's derivative is

$$g_1^\top d_\alpha = 2(3\alpha - 1) > 0 \quad \Longleftrightarrow \quad \alpha > \frac{1}{3}.$$

The second property's derivative is

$$g_2^\top d_\alpha = -(3\alpha - 1) + (1 - \alpha) = 2 - 4\alpha > 0 \quad \Longleftrightarrow \quad \alpha < \frac{1}{2}.$$

Their intersection is

$$\boxed{\alpha \in \left(\frac{1}{3}, \frac{1}{2}\right)}.$$

At either endpoint one derivative is zero. The conclusion concerns a sufficiently small positive step at the specified state. Gradients can change later in the trajectory.

Step 5 continued

(c) Some users specify two desired attributes through classifiers. Adding the two classifier gradients may overlook dependence between those attributes. Derive the missing correction before interpreting the combined guidance as conditioning on both attributes.

At a fixed time and observation y , let

$$\pi_i(z) = \Pr(C_i \mid Z = z, y), \quad \pi_{12}(z) = \Pr(C_1 \cap C_2 \mid Z = z, y),$$

with all probabilities positive. Let $s = \nabla_z \log q(z \mid y)$. An implementation uses $s_{\text{add}} = s + \nabla \log \pi_1 + \nabla \log \pi_2$. Use Bayes' rule to derive the correction $\Gamma(z)$ such that

$$s_{\text{add}}(z) + \Gamma(z) = \nabla_z \log q(z \mid C_1 \cap C_2, y).$$

Box one equation for Γ using the three classifier probabilities.

(0.5 points)

Bayes' rule for the joint event gives

$$q(z \mid C_1 \cap C_2, y) = \frac{q(z \mid y)\pi_{12}(z)}{\Pr(C_1 \cap C_2 \mid y)}.$$

The denominator is independent of z , so the joint conditional score is $s + \nabla \log \pi_{12}$. Subtract the implemented score:

$$\begin{aligned} \Gamma &= \nabla \log \pi_{12} - \nabla \log \pi_1 - \nabla \log \pi_2 \\ &= \nabla \log \left(\frac{\pi_{12}}{\pi_1 \pi_2} \right). \end{aligned}$$

Thus

$$\Gamma(z) = \nabla_z \log \frac{\pi_{12}(z)}{\pi_1(z)\pi_2(z)}.$$

Conditional independence given (z, y) makes the ratio one and the correction zero. More generally, a ratio constant in z also has zero gradient.

Step 6: Interpreting Benchmarks and Ablations (2.5 points)

The paper must evaluate the properties established by the mathematics and the accuracy of the learned conditional distribution. All results in this step are hypothetical, with exact values for this exercise. Training budgets, held-out observations, candidate counts, and sampling costs are matched unless a change is explicitly stated. Lower errors are better.

In a controlled benchmark, the target conditional means and covariances are known. Define the moment error as

$$E_{\text{mom}} = \|\hat{\mu} - \mu_{\text{target}}\|^2 + \|\hat{\Sigma} - \Sigma_{\text{target}}\|_F^2,$$

where $\|\cdot\|_F$ is the square root of the sum of squared matrix entries. Measurement error is $E_{\text{obs}} = \mathbb{E}\|AX - y\|^2$.

Method	E_{obs}	E_{mom}
Unconditional generator, final projection	0	0.20
Conditional flow in free coordinates	0	0.05
Conditional diffusion in free coordinates	0	0.04

(a) Use both measurements to assess the paper's central claim. The projection theorem guarantees agreement with the observations, while the moment comparison tests additional information about the distribution of completions.

Which conclusion is supported? Choose ONE.

1. Zero measurement error establishes exact conditional sampling for all three methods.
2. The two conditional methods have smaller moment error in this benchmark. Zero measurement error alone does not establish conditional fidelity.
3. Diffusion is more accurate than every flow method on every conditional generation problem.
4. A small moment error proves equality of the complete generated and target distributions.

(0.5 points)

All three methods satisfy the measurement constraint. Their conditional moment errors differ, so feasibility alone cannot establish the desired conditional law. The conditional methods have smaller measured moment errors on this benchmark.

Mean and covariance also do not determine a general distribution, so these measurements cannot certify full distributional equality. The supported conclusion is

Step 6 continued

(b) A small collection of score checks can miss a structured error. Use an analytic ablation to determine which additional probe would detect an error hidden from simple low-order checks.

Let $q = \mathcal{N}(0, 1)$ and suppose a perturbed score is

$$s_\delta(z) = -z + \delta(z^3 - 3z), \quad \delta < 0.$$

Use the residual from Step 3(e). Polynomial test functions are allowed here because Gaussian decay makes the boundary terms vanish. You may use $\mathbb{E}[Z^2] = 1$, $\mathbb{E}[Z^4] = 3$, $\mathbb{E}[Z^6] = 15$, and zero odd moments.

Which ONE test function has a nonzero population residual and therefore detects this perturbation?

1. $h(z) = 1$.
2. $h(z) = z$.
3. $h(z) = z^2$.
4. $h(z) = z^3$.

Box only the option number.

(0.5 points)

The true score is $s(z) = -z$. By Step 3(e),

$$\mathcal{R}_\delta(h) = \delta \mathbb{E}[(Z^3 - 3Z)h(Z)].$$

Evaluate the four population residuals:

$$\begin{aligned} \mathcal{R}_\delta(1) &= 0, \\ \mathcal{R}_\delta(z) &= \delta(3 - 3) = 0, \\ \mathcal{R}_\delta(z^2) &= 0, \\ \mathcal{R}_\delta(z^3) &= \delta(15 - 9) = 6\delta \neq 0. \end{aligned}$$

The first and third vanish by symmetry. The cubic probe detects the error missed by the other three, so

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This example demonstrates why finitely many zero diagnostic residuals do not establish an exact score.

Step 6 continued

To check whether coordinate scaling changes the intended diffusion, repeat a reference run after $z = Ru$ with $R = \text{diag}(2, 1)$. Transform the drift and initialization consistently. The only ablated choice is the free-coordinate covariance C_U of the Gaussian increments, after removing their common factor g^2h .

Run	Noise covariance in u	E_{mom}
Reference in z	I_2 in z	0.01
Rescaled, covariance corrected	$R^{-1}R^{-\top}$	0.01
Rescaled, covariance correction removed	I_2	0.18

(c) Interpret this ablation through the covariance transformation from Step 4(f). A change of coordinates should preserve the physical diffusion when every coefficient is transformed consistently.

Which mechanism explains why the last run no longer implements the reference SDE? Choose ONE.

1. Its diffusion covariance in z is $R^{-1}R^{-\top}$, which reduces noise in the first direction.
2. Its initialization must change even though the stated protocol transformed it consistently.
3. Its diffusion covariance in z is $RR^{\top} = \text{diag}(4, 1)$, so the first free direction receives four times the reference noise variance.
4. Rescaling coordinates necessarily changes the physical process even with the covariance correction.

(0.5 points)

The ablated increment is $\Delta U = g\sqrt{h}\xi$ with covariance g^2hI_2 . Under $\Delta Z = R\Delta U$,

$$\text{Cov}(\Delta Z) = g^2hRR^{\top} = g^2h \text{diag}(4, 1).$$

Thus the diffusion coefficient has changed in the physical free coordinates. The corrected run instead has

$$RC_UR^{\top} = R(R^{-1}R^{-\top})R^{\top} = I_2.$$

The correct selection is

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The table is consistent with the mathematical change in the ablated SDE. It does not establish a universal quantitative relation between covariance error and moment error.

Step 6 continued

(d) Removing projection can increase reward by changing the known measurements. Quantify this violation before interpreting the ablation as an improvement.

Start from $Ax = y$ and take $x^+ = x + h\eta g$, with fixed full-state gradient $g \in \mathbb{R}^d$, $h > 0$, and $\eta \geq 0$. Box the resulting squared measurement error $\|Ax^+ - y\|^2$. (0.5 points)

The initial constraint gives $Ax^+ - y = h\eta Ag$. Taking the squared Euclidean norm yields

$$\|Ax^+ - y\|^2 = h^2\eta^2\|Ag\|^2.$$

With Pg in the update, $AP = 0$ makes this error zero. The ablation can change the known measurements whenever $Ag \neq 0$.

(e) After checking feasibility, compare guidance with independent property measurements. A high predictor score can reflect prediction error, so its gradient may favor changes that reduce the properties you care about.

All runs below use projected guidance and have zero measurement error. Both independent property scores should increase.

Strength η	Guiding predictor	Q_1	Q_2
0	0.50	0.50	0.50
1	0.75	0.68	0.64
3	0.95	0.56	0.39

Which next step is best supported? Choose ONE.

1. Select $\eta = 3$ because its guiding-predictor score is highest.
2. Infer that feasibility guarantees improvement in both independent properties.
3. Use validation data to test guidance strength and predictor reliability, then report both properties on an untouched test set at matched cost.
4. Remove projection while claiming the original constraint guarantee.

(0.5 points)

Increasing η from 1 to 3 raised the guiding score from 0.75 to 0.95, while Q_1 fell from 0.68 to 0.56 and Q_2 from 0.64 to 0.39. Larger predictor scores therefore did not imply better independent properties.

Select guidance using validation data, then compare both properties on an untouched test set with feasibility and cost controlled. Thus

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End of practice exam.

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